METHOD FOR COLISTIN FOR 2018 MAY BE REVISED IN THE NEXT 4 MONTHS

MARY K. YORK, PH.D.

VLADEMIR CANTARELLI

AMERICAN SOCIETY FOR MICROBIOLOGY

LEARNING OBJECTIVES

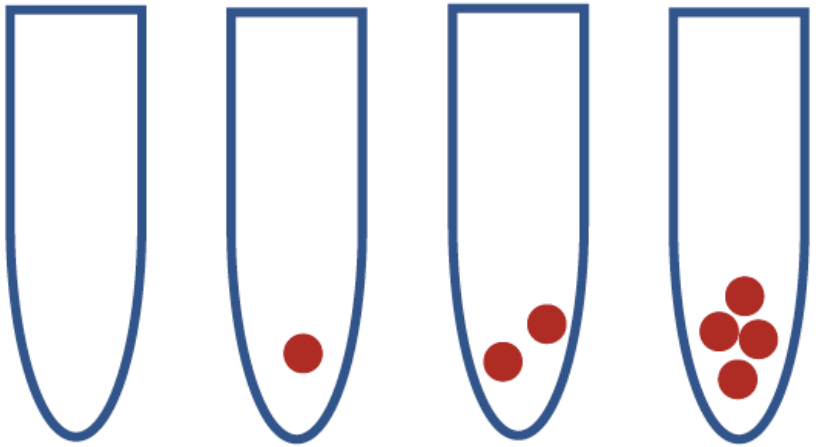
At the end of the lecture, participants should be able to:

- Demonstrate technique for detection of colistin resistance
- Understand the way CLSI works
- Know the importance of checking new M100 every Janaury

AMERICAN SOCIETY FOR MICROBIOLOGY

Method

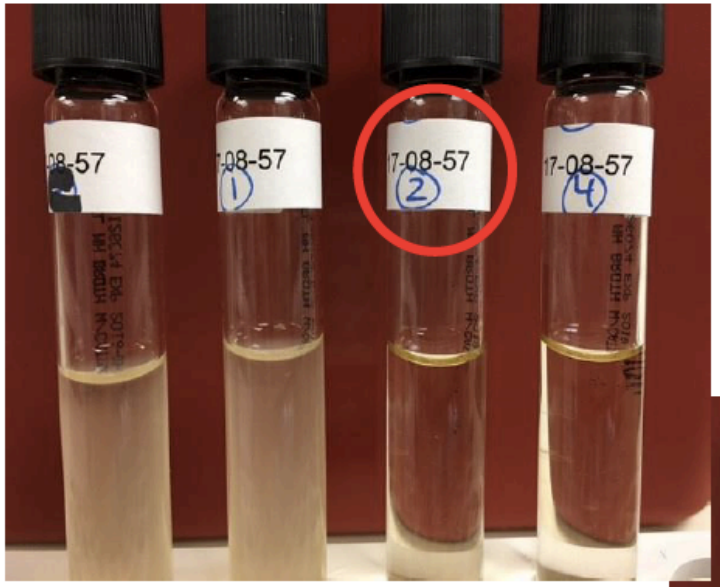
Broth Disk Elution Method



- Add 1, 2, or 4 colistin 10 µg disks to 10 ml CAMHB
- Add 50 µl McF 0.5 inoculum
- Incubate overnight

AMERICAN SOCIETY FOR MICROBIOLOGY

Reading



CDC mcr-1 (AR 349)
Published result = 2-4



CDC mcr-1 (AR 493)
Published result = 8

AMERICAN SOCIETY FOR MICROBIOLOGY

Data

Colistin Broth Disk Observations
UCLA Isolates

Enterobacteriaceae

- Clear endpoints
- 3/6 mcr-1 isolates MIC = 2 (BMD for 2/3 = 4)

Acinetobacter spp.

- Must incubate 24 h
- Some slight hazes - above the MIC? (what to do with these?)

P. aeruginosa

- 18 h agrees with BMD
- 24 h hazes and MIC > BMD

AMERICAN SOCIETY FOR MICROBIOLOGY

Not yet published

- Needs more testing in 3 labs
- Needs vote of subcommittee
- Should be in M100 S28
- Where
  - In the list of changes in the beginning
  - ??? In Table 3 after mCIM

AMERICAN SOCIETY FOR MICROBIOLOGY

From trip tp Delta – not from diving



AMERICAN SOCIETY FOR MICROBIOLOGY